

Input Parameters

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

Ben Geisler

2026-02-23

Table of contents

1	Summary	1
2	Global Model Parameters	1
3	Model Input Parameters	2
4	PSA Parameter Distributions	3

1 Summary

This report provides an overview of all input parameters used in the METIMMOX-1 health economic model, organized into global model settings and variable parameters across base case, DSA, and PSA analyses.

2 Global Model Parameters

These parameters define the overall model structure and are held constant across all analyses.

Table 1: Global Model Parameters

Parameter	Value	Description	Source
Time Horizon	520 weeks (10 years)	Model time horizon	Clinical follow-up period
Cycle Length	1 week	Cycle length for discrete-time model	Model assumption
Discount Rate	4% per annum	Annual discount rate for costs and effects	Model specification
WTP Threshold	€51,000 per QALY	Willingness-to-pay threshold	Norwegian threshold based on severity (14.5 QALYs lost for diagnosis, based on a health 65-year old)
PSA Sample Size	5,000 iterations	Number of Monte Carlo iterations for PSA	Computational adequacy
Sampling Models	5,000 models	Resampled models for survival uncertainty	Statistical adequacy
DSA Variation Range	$\pm 20\%$	Parameter variation range for DSA	Standard practice
Model Structure	0 (joint)	Survival model formula: 0=joint (all biomarkers + all interactions), 1=focused (per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction), 2=separate (one biomarker only)	Methodological choice

3 Model Input Parameters

This table presents all variable model parameters organized by type, showing their base case values, DSA variation ranges, and PSA distribution specifications.

Table 2: Model Input Parameters by Type

Type	Parameter	Base Case	DSA Range	PSA Distribution
Cost Parameters				
Cost	Nivolumab per administration	€13,923	€11,138 - €16,708	Gamma (shape=25, rate=0.0018)
Cost	FLOX per administration	€427	€342 - €512	Gamma (shape=25, rate=0.0586)
Cost	CT scan	€386	€309 - €463	Gamma (shape=25, rate=0.0648)
Cost	Blood tests	€0	Not varied	Fixed at 0
Cost	NGS testing	€1,439	€1,151 - €1,727	Gamma (shape=25, rate=0.0174)
Cost	Standard outpatient visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	Baseline visit	€530	Not varied	Gamma (shape=25, rate=0.0472)
Cost	Follow-up visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	End-of-life care	€13,803	€11,042 - €16,564	Gamma (shape=25, rate=0.0018)
Utility Parameters				
Utility	Non-progressed state	0.9077	0.7262 - 1.0000	Beta (=41.0, =4.2)
Utility	Progressed state	0.9005	0.7204 - 1.0000	Beta (=40.6, =4.5)
Prevalence Parameters				
Prevalence	CRP biomarker positive	30%	24% - 36%	Beta (CV=15%)
Prevalence	TLR biomarker positive	25%	20% - 30%	Beta (CV=15%)
Prevalence	TMB/BRAF biomarker positive	20%	16% - 24%	Beta (CV=15%)
Survival Parameters				
Survival	OS and PFS curves by biomarker status	Fitted parametric survival models	Not varied	Survival resampling (5,000 models)

Notes:

DSA ranges calculated as base case $\pm$ 20%. Utility upper bounds capped at 1.0.

PSA: Gamma distributions for costs (CV=20%), Beta for utilities (CV=15%).

Survival curves use non-parametric resampling with age and sex adjustments.

4 PSA Parameter Distributions

This section displays histograms of all sampled parameters from the probabilistic sensitivity analysis (PSA), showing the distribution shapes and ranges used in the Monte Carlo simulation.

PSA parameter data not available.

To generate these histograms:

1. Run your PSA analysis script (e.g., scripts/R/analysis/13_EVPPIs.R)
2. Save the psa_params object using: `saveRDS(psa_params, here('results', 'psa_params.rds'))`
3. Re-render this report

Report completed on: 2026-02-23

Repository: ben-geisler/METIMMOX-1

Report version: 2.1